

Data Flow Diagram - LegalEase System

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Project Name: LegalEase: AI-Powered Legal Document Generator **Maximum**
Marks: 2 Marks

1. Data Flow Diagram Overview

This Data Flow Diagram (DFD) illustrates how data moves through the **LegalEase** system—between external entities, backend processes, data stores, and AI services.

2. DFD Symbol Legend

Symbol Name	Description
Oval / Rounded Shape	External Entity: A person, organization, or external system outside the project that sends or receives data (e.g., User / Streamlit Interface, Gemini 1.5 Pro API).
Rectangle with Numbered Header	Process: An activity that transforms incoming data into outgoing data (e.g., Process Input, Generate Prompt, Format Output).
Rectangle (Solid Fill, No Number)	Data Store: A place where data is held for later use (e.g., Environment File / API Key Config, Document History Data Store).

Labeled Arrow	Data Flow: The movement of data between entities, processes, and data stores, labeled
Symbol Name	Description
	with what data is being passed.

3. System Process Breakdown

Process 1: User Request & Frontend Collection (Streamlit)

- **External Entity:** User (via Streamlit Interface)
- **Input Data:** Document Type (e.g., Contract, NDA, Lease), Parties Involved, Terms & Conditions, Effective Date.
- **Process (1.0):** Validate Input Data & Send POST Request via app.py to FastAPI Backend.

Process 2: FastAPI Request Handling & Prompt Engineering

- **Input Data:** JSON Payload (DocumentRequest model containing document attributes).
- **Process (2.0):** Parse JSON in routes.py and build structured prompt inside ai_core/gemini_generator.py.
- **Data Store Access:** Reads API keys from Configuration / .env file.

Process 3: AI Document Generation (Gemini 1.5 Pro)

- **External Entity:** Gemini 1.5 Pro API (Google Generative AI)
- **Process (3.0):** Transmit structured prompt; receive generated legal text response.
- **Output Data:** Raw legal text response with clauses and structured metadata.

Process 4: Output Formatting & Export Handling

- **Process (4.0):** Sanitize text and format into user-requested document formats via format_docx(), format_pdf(), and format_html_preview().
- **Output Entity:** User Interface preview & downloadable file export buttons (.TXT, .DOCX, .PDF).

4. Data Flow Architecture Mapping

